

A plausible overlap that fails the test

OHRC to TMC-2 | Adjusted-camera audit | Real science data

**REJECTED -
CONSISTENCY**

The footprints overlap, but image-derived displacement and independent terrain geometry disagree. Accepted science control points: zero.

1 / What was tested

The original OHRC science image and TMC-2 ortho were evaluated with an adjusted OHRC camera, real TMC DTM and independent ASP terrain alignment. This is the earlier candidate, separate from the later REALPAIR005 local registration.

Instrument	Exact product ID	Role
OHRC	ch2_ohr_nrp_20200827T0030107497_d_img_d18	Optical source
TMC-2	ch2_tmc_ndn_20231101T0125121377_d_oth_d18	5 m ortho target
TMC-2 DTM	ch2_tmc_ndn_20231101T0125121377_d_dtm_d18	Terrain reference

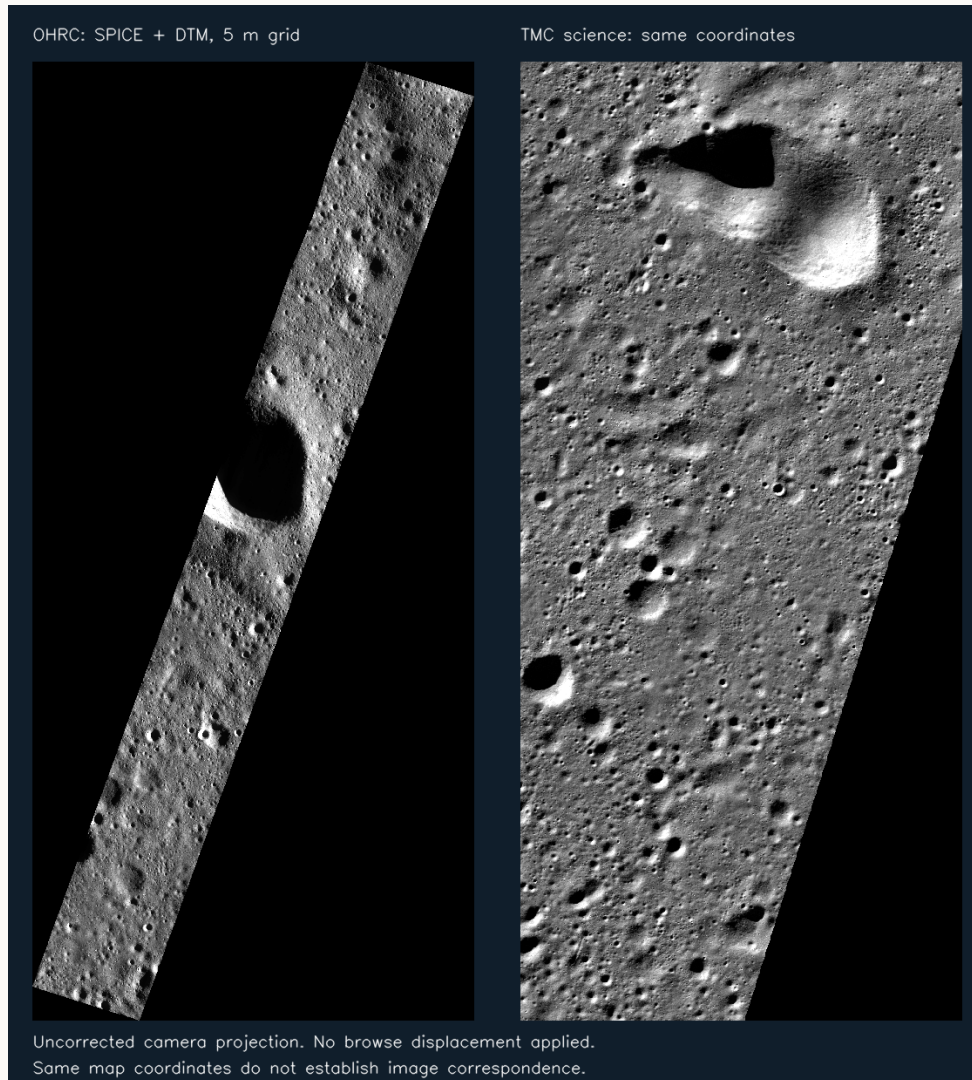
2 / Why it is rejected

Gate	Measured result	Decision
Camera / DTM	ISIS-CSM maximum round trip 0.014 OHRC px; stereo adjustment 10,690 points	Internal geometry passes
SIFT support	8/19 (42.11%) training; 0/6 (0.00%) held-out; 5/18 cells	Fails held-out and spread
AKAZE support	36/85 (42.35%) training; 25/50 (50.00%) held-out	Point ratio passes
AKAZE coverage	8/18 cells; 9.3% hull versus 15% gate	Fails spatial coverage
Independent geometry	6.97 km image shift versus 0.65 km ASP terrain estimate	Conflict

The ~6.97 km image shift remains a hypothesis. It was not applied as a camera correction or accepted as a control network. SIFT and AKAZE can agree on repeated crater texture while still selecting the wrong physical region.

A shared map frame is not a match

The images below are real science-data context from the earlier camera audit. They show where the camera projected OHRC and the TMC ortho lie on the same 5 m map grid; they do not certify pixel correspondence.



Earlier projection comparison only. The final v5 rejection uses an adjusted camera and frozen matcher/terrain metrics, reported above.

3 / Evidence boundary

Stereo bundle adjustment and camera-model equivalence validate internal OHRC geometry, not its absolute tie to TMC-2. Illumination and shadows make this pair difficult, but the audit does not identify them as the sole cause. The independent ASP alignment of 10,690 OHRC terrain points to the TMC DTM estimated 649.64 m translation and 97.88 m output RMSE; those are terrain-alignment results, not accepted image control points.

4 / Reproducibility

Audit ID: **REALPAIR001_adjusted_registration_v5**. audit_summary.json records the decision. The SIFT and AKAZE experiment folders contain frozen config, matches, inliers, reverse matches, transform, metrics and manifest files. pc_align/ retains independent terrain results; checksums.sha256 binds the critical artifacts. The next valid action is a new illumination-compatible pair or independent lunar control with untouched validation data.

Source: data/derived/REALPAIR001_adjusted_registration_v5/SCIENTIFIC_REPORT.md